6 **(a)** A student claims that "Light is only a wave and not a particle."

Discuss whether this statement is correct, providing experimental evidence.

When light passes through two closely spaced slits, bright and dark fringes show the constructive and destructive interference of the light waves. In this case light behaves as waves. [1]

When light above a certain threshold frequency is shone on a metal surface, electrons are emitted from the metal surface. This provides evidence that supports the particulate nature of light. [1]

Therefore, the statement is incorrect. Light is both a wave and a particle. [1]

[3]

(b) A cool, low-pressure gas of an element X is illuminated by a beam of monochromatic violet light of frequency 8.21×10^{14} Hz.

The simplified energy level diagram for an atom of element X is shown in Fig. 6.1 below.

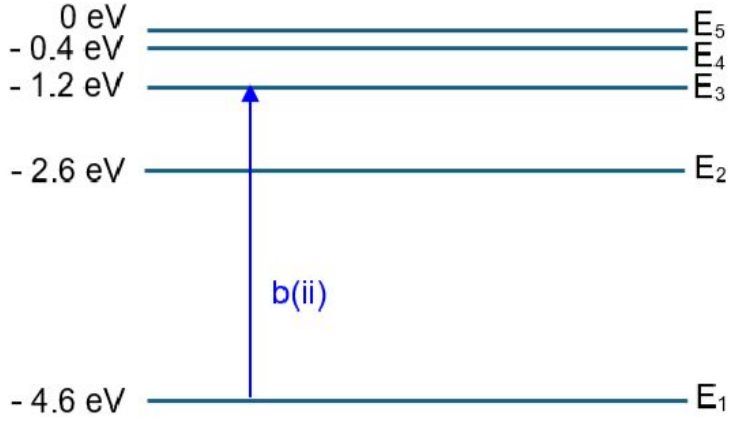


Fig. 6.1

(i) Calculate the energy of a photon of the incident violet light.

$E = hf$

$E = (6.63\times10^{-34})(8.21\times10^{14})/1.6\times10^{-19}$ [1]

$E = 3.4$ eV [1]

Energy of a photon = ... eV [2]

(ii) Draw, on Fig. 6.1, the transition corresponding to the absorption of the incident violet photon. [1]

(iii) Determine the number of different emission lines that may be observed as a result of the absorption of many of such violet photons.

Number of emission lines = ... 3 ..... [1]

(iv) Explain why the intensity of the violet light decreases after passing through the gas.

As the violet light photons are absorbed by the atoms in the cool gas, exciting the electrons from ground state to the higher energy level E_3. [1]

When these excited electrons de-excite, photons of different energies are emitted in **all random directions** rather than continuing purely along the original direction of the beam. The number of violet photons which travel straight through is lower than before, [1] hence lower intensity of violet light is observed. [2]

[Total: 9]